

Photodynamic Therapy (PDT) as an Oncological Treatment Modality

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PHYS 383 – Medical Physics

February 15, 2024

Introduction

Overview

Photodynamic therapy (PDT) is an innovative and promising approach in the realm of cancer treatment that uses light and photosensitizing agents to target and eliminate cancerous cells selectively. These agents are absorbed into the cancer cells then act as targets during treatment, and release unstable oxygen which decays producing heat to kill cells upon activation. Unlike treatment modalities like as surgery, chemotherapy, or radiation therapy, PDT uses light to activate photosensitizers accumulating in malignant tissues. In essence, PDT represents a paradigm shift in cancer therapy, harnessing photochemistry for precise and targeted interventions in the fight against cancer. This article explores the intricacies of PDT, covering its mechanism of action, applications in cancer treatment, and the promising avenues it opens for the future.

Main Content

Mechanism of Action

Photodynamic therapy relies on the physics of light interaction with photosensitizing chemicals (i.e., photosensitizers, photosensitizing agents) to destroy targeted cells. Photosensitizers, such as porphyrins and chlorins, absorb light at specific wavelengths, causing photoexcitation and boosting the molecules' energy levels [4]. The subsequent transfer of energy to molecular oxygen produces reactive oxygen species (ROS), which are superoxide and hydroxyl radicals formed in Type I reactions and singlet oxygen in a Type II reaction [1]. These ROS cause damage to biological components, resulting in apoptosis or necrosis in the target cells [2]. When administered systemically, photosensitizers selectively accumulate in target tissues,

particularly cancer cells. To match the absorption spectra of photosensitizers, the light wavelength and source, which are frequently lasers or LEDs, must be carefully considered [2].

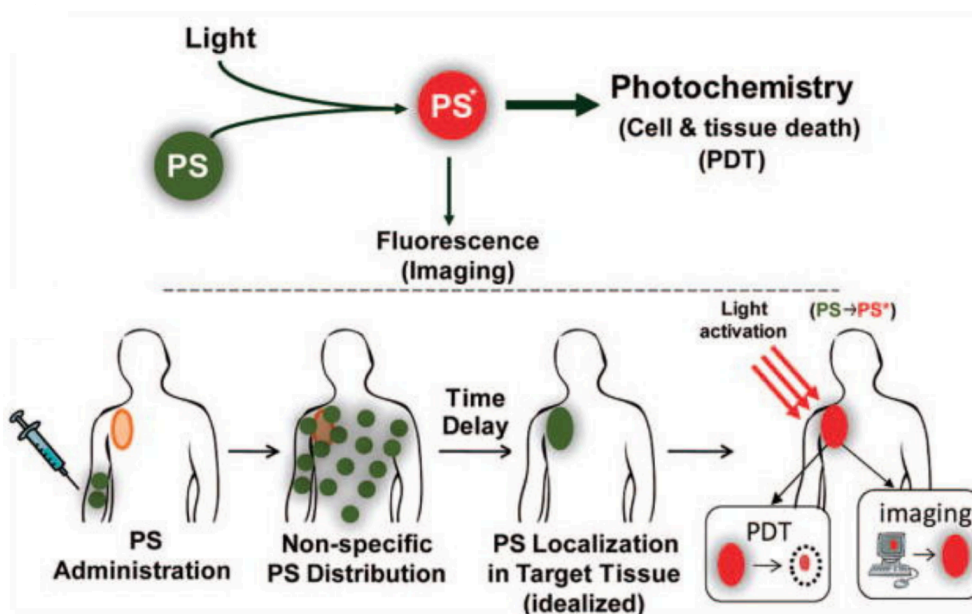


Figure 1. A graphic illustration summarizing the photochemical action mechanism of photodynamic therapy [4]

As light determines PDT efficacy, the depth of tissue penetration is affected and leads to the choice of suitable settings [5]. Understanding these physical complexities is crucial for optimizing PDT, directing decisions about photosensitizers, light dosages, and overall treatment strategies, and contributing to its future development and application in various medical contexts.

Photosensitizers and Photosensitizing Agents

Central to PDT are photosensitizers, which are compounds that, when exposed to specific wavelengths of light, generate reactive oxygen species (ROS) causing cellular damage and ultimately cell death. Common photosensitizers include porphyrins, chlorins, and phthalocyanines [1]. Each of these compounds has unique absorption spectra, influencing the choice of light source for activation. They can be intravenous, orally ingested, or applied on top of the skin [1].

Application in Cancer Treatment

Common Cancer Types

PDT has found application in treating tumours located either on or just beneath the skin, or on the lining of internal organs or cavities—bodily surfaces which are reachable by visible light. Carcinoma, which is a cancer that develops in epithelial tissue, is the most prevalent kind of cancer treated with PDT [6]. Some examples are skin cancer (e.g., Basal Cell Carcinoma and Squamous Cell Carcinoma), early-stage lung cancer and throat or esophageal cancer [12][7][9]. There is a difference in how PDT is administered or how cancer cells are targeted between these various cancer types. For skin tumours, the light is targeted right at the tumour itself while for lung and throat cancers, an endoscope is used to reach the tumour inside the body along with a fibre optic cable which transmits the visible light to the target [11].

Advantages and Limitations

PDT has various advantages over other treatment techniques, the first of which is its localized approach. This requires strategically placing photosensitizers within tumour lesions, which are then precisely irradiated [5]. This targeted application makes PDT very selective against tumour cells. Another advantage is that PDT is minimally invasive, making it a less intrusive option to surgical therapies, and it can be provided as an outpatient under a pre-determined treatment plan [5]. Additionally, PDT can be repeated and combined with other modalities such as surgery, chemotherapy, and radiotherapy, thereby enhancing overall clinical efficacy—a facet that will be further explored in the subsequent discussion on PDT's multimodal potential as a future avenue in cancer treatment [5].

Despite the outlined advantages, the primary constraint facing PDT is tissue oxygenation. Because the photodynamic effect is dependent on tissue oxygenation, tumours with thick tumour

masses or those surrounded by necrotic tissue may not react to photodynamic therapy [5]. Another constraint is post-treatment photosensitivity [9]. Porfimer sodium, a first-generation photosensitizer, causes photosensitivity in the skin and eyes for around six weeks [9]. Due to the induced photosensitivity, care should be taken to avoid direct sunlight and/or intense interior lighting during this time. Along with tissue oxygenation and photosensitivity, there is limited tissue penetration with visible light [3]. PDT's therapeutic success is determined by its accuracy of delivery to the target tissue and dosage, therefore deep tumours that cannot be accessed without surgery present a difficulty. Recognizing the challenge of limited tissue penetration, researchers have now investigated nanoparticle-based photosensitizer delivery systems for PDT, intending to improve its specificity, or the ability to target cancer cells precisely.

Future Avenues

Nanoparticle-based Drug Delivery System

Nanoparticle-based drug delivery systems offer the potential to improve the targeted delivery of photosensitizers to cancer cells, minimizing collateral damage to healthy tissues. Simultaneously, gold, renowned for its multifaceted medicinal properties, has evolved as a key player in modern medicine. Gold nanoparticles are good photosensitizing agents due to their bacteriostatic, anticorrosive & antioxidative properties; they also exhibit localized plasmon surface resonance (LPSR), which allows them to be used at a variety of light wavelengths [12]. Gold, in its nanoscale form, can passively traverse the body, selectively accumulating in tumours with leaky blood arteries before being safely eliminated through the urine system [12].

Real-time Imaging and Monitoring

Real-time monitoring tools are currently lacking in the PDT landscape; thus, including real-time imaging and monitoring technology in PDT operations could provide essential insights

into treatment success. Monitoring the dispersion and activation of photosensitizers in real time could help clinicians optimize therapy parameters [10]. Specifically, synchronous feedback can enable quick modifications, improving therapy efficacy and reducing unnecessary injury to healthy tissues [10].

Multimodal Therapy

Combining PDT with other cancer treatment modalities, such as chemotherapy or radiotherapy, opens up new directions for enhanced therapeutic outcomes. Strategically integrating these modalities with personalized dosimetry and treatment plans suited to unique patient characteristics can increase treatment efficacy, mitigate the risk of tumour drug resistance, and reduce treatment duration [8].

Conclusion

Photodynamic therapy (PDT) represents a cutting-edge paradigm in cancer treatment, offering a unique blend of precision and minimally invasive intervention. From its photophysical mechanism to its applications, advantages, and limitations, PDT is shaping the future of radiation oncology. As industries embrace this technology and researchers explore new directions, the potential for even more effective cancer treatments continues to brighten. With the ongoing evolution of nanoparticle-based drug delivery, real-time imaging, and multimodal therapy, PDT is poised to play a pivotal role in the multifaceted landscape of cancer care. As we continue to unravel the intricacies of PDT, its potential impact on personalized and targeted cancer treatments becomes increasingly apparent, marking a hopeful era in the fight against cancer.

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